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REPORT WRITING

ASSIGNMENT NO # 02

Project Title: Silent Classroom (Real-Time Confusion Detector)

- **App Idea:** A web application that allows students to anonymously signal their level of understanding (e.g., "Got it", "Following", "Lost") during a lecture. Teachers see this feedback live on a dashboard with charts and a live activity feed, instead of asking "any questions?" and getting silence.
- **Purpose of the Project:** To bridge the communication gap between students and teachers in a classroom setting. Many students hesitate to admit confusion publicly. This tool gives teachers real-time, anonymous insight into class comprehension so they can adjust their pace or re-explain topics immediately.
- **AI Tools Used:** Claude (for code generation, debugging help, and deployment guidance).

- **Initial Prompt Used:** A prompt describing the concept an app where students can anonymously click buttons to indicate confusion/understanding, and a teacher dashboard that displays this data live using charts.

- **Prompt Iterations During Development:**

- First iteration generated the basic UI structure (student view + teacher dashboard).
- Follow-up prompts asked to add a live feed of signals, charts (using Recharts) to visualize "Got it / Following / Lost" trends over time, and a "Join Session" feature.
- Later prompts focused on refining the layout and styling for clarity.

- **What Worked Well:**

- Claude generated a functional React + Vite project structure quickly.
- Component-based design (separate App.jsx, main.jsx) made the code easy to understand.
- Integration of Recharts for live visual feedback worked smoothly once dependencies were correctly set up.

- **What Did Not Work:**

- Initial file structure was not deployment-ready (App.jsx and main.jsx were in the wrong folder).
- A misnamed config file (vite_config.js instead of vite.config.js) caused build failures.

- **Bugs/Challenges Faced:**

- Vercel build failed with "Command npm run build exited with 1" due to incorrect file paths (index.html referenced /src/main.jsx, but the files were in the root directory).
- Confusion around uploading project files to GitHub (zip vs. extracted files, excluding node_modules).

- **How the Issues Were Solved:**

- Created a proper src/ folder in the GitHub repository and moved App.jsx and main.jsx into it, matching the path expected by index.html.
- Renamed the config file to vite.config.js.
- Excluded node_modules from the GitHub upload since Vercel installs dependencies automatically during build.
- Re-deployed on Vercel, which then built successfully.

➤ **Features Added or Improved:**

- Anonymous student feedback buttons (Got it / Following / Lost).
- Real-time line chart visualization of class understanding trends.
- Live feed section showing incoming signals.
- "Join Session" functionality for students to enter a classroom session.

➤ **Final Result of the Project:** A fully functional, deployed web application accessible via a public link (silent-classroom-app.vercel.app), built with React, Vite, and Recharts, hosted on Vercel with continuous deployment from GitHub.

➤ **Does the Project Solve a Real-World Problem?**

Yes, it addresses a common classroom issue where students are reluctant to voice confusion, helping teachers identify learning gaps in real time and improving overall teaching effectiveness.